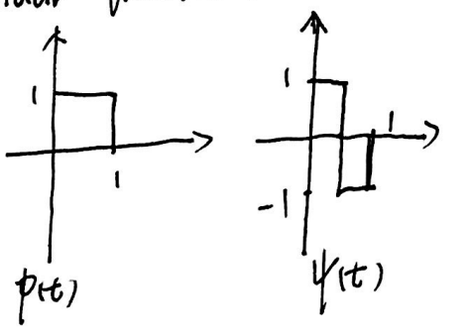
Haar function.



可能会考 How $\phi(t)$ and $\psi(t)$ be represented by combination of $\phi(2t)$ and $\phi(2t-1)$

$$\phi(t) = \phi(2t) + \phi(2t-1)$$
$$\psi(t) = \phi(2t) - \phi(2t-1)$$

↑ 诸如此类的可能会有所变化

The Wavelet Transform.

Works on nonstationary data.

useful in many types of applications: Pattern recognition
- Feature extraction
- Trend detection
- Perfect reconstruction
- Video compression

CWT (Continuous Wavelet Transform)

$$CWT(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} s(t)\, \psi^*\left(\frac{t-b}{a}\right) dt = \int_{-\infty}^{\infty} s(t)\, \psi_{a,b}^*(t)\, dt = \langle s, \psi_{a,b} \rangle$$

CWT(a,b) is a function of two real variables.

=

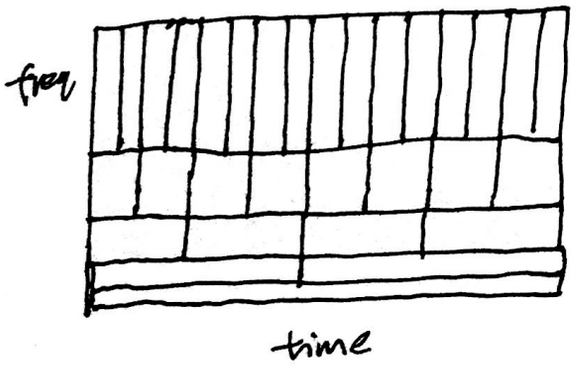
CWT provides a time-frequency as well as time-scale representation.

联系 CWT & STFT

CWT: Variable time-frequency resolution

STFT: Constant time-frequency resolution.

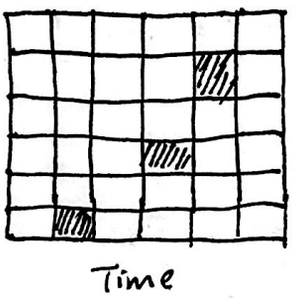
Partition of the time-frequency plane



High scale (low frequency)
— large window size. better frequency resolution

Low scale (high frequenay)
— small window size, better time resolution.

Time-Freq Partition STFT



FT: Equal time and frequency resolution.

WT: Logarithmic scale of frequency resolution.

Inverse CWT: The Admissability Criterion.

若要 inverse CWT, 要滿足 Admissibility condition

- square of the Fourier Transform must decay faster than $\frac{1}{\omega}$
- Admissibility is measure of signal's band-limitedness
- Admissibility implies zero average.

比較 STFT & CWT.

Similarity:
- signal is multiplied by a function, and the transform is computed separately for different segments of signals.
- Can be written in inner product form
- Time-frequency window area remains constant.

Difference:
- Fixed time duration and freq bandwidths of $\gamma(t)$ [(STFT)]
- Variable time duration and bandwidth of $\psi(t)$ [(CWT)]

Problems with CWT:
- Redundancy
- Infinite solution space
- Efficiency

：具体解释在 PPT WT那一章的16页

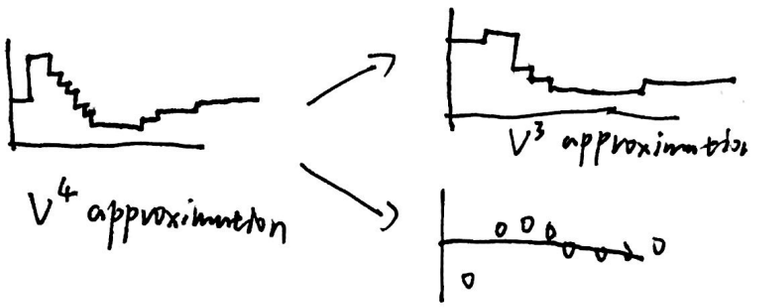
Multiresolution

Piecewise Approximation.

Basic Concept: Decompose a fine-resolution signal into

① A coarse-resolution version of the signal.

② The differences left over



左侧加上课堂的例子.

n=3 [56, 40, 8, 24, 48, 48, 40, 16]

n=2 [48, 16, 48, 28, 8, -8, 0, 12]

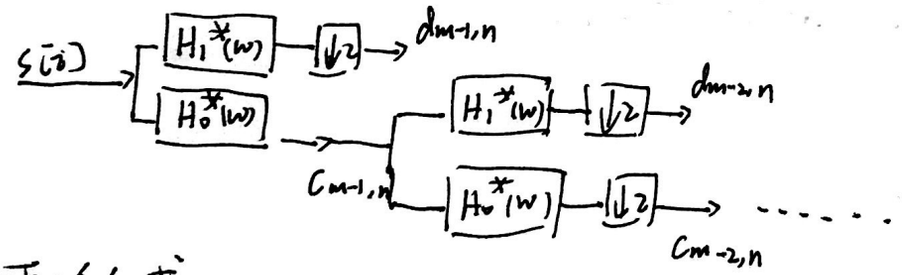
n=1 [32, 38, 16, 10, 8, -8, 0, 12]

n=0 [35, -3, 16, 10, 8, -8, 0, 12]

Wavelet Transform from Filter Bank.

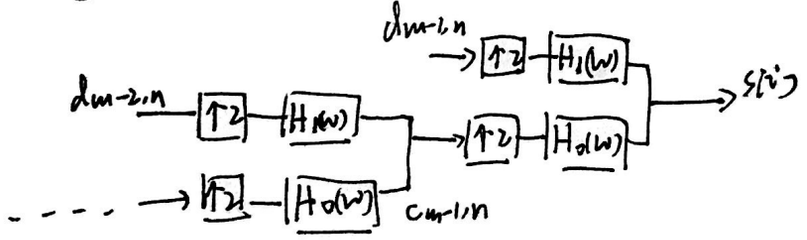
$$C_{m-1,n} = \sqrt{2} \sum_i h_0[i-2n]\, C_{m,i}$$

$$d_{m-1,n} = \sqrt{2} \sum_i h_1[i-2n]\, C_{m,i}$$



也可以反向

$$C_{m,n} = \sqrt{2}\left[\sum_i h_0[n-2i]\, C_{m-1,i} + \sum_i h_1[n-2i]\, d_{m-1,i}\right]$$



Daubechies Wavelet family

$H_0[n] = [H_0[0], H_0[1], H_0[2], H_0[3]]$

$H_1[n] = [H_0[3], -H_0[2], H_0[1], -H_0[0]]$

$\gamma_0[n] = [H_0[2], H_0[3], H_0[1], H_0[0]]$

$\gamma_1[n] = [-H_0[0], H_0[1], -H_0[2], H_0[3]]$

eg1:

n=3 [10, 14, 26, 24, 30, 20, 8, 16]

n=2 [12, 25, 25, 12, -2, 1, 5, -4]

n=1 [18.5, 18.5, -6.5, 6.5, -2, 1, 5, -4]

n=0 [18.5, 0, -6.5, 6.5, -2, 1, 5, -4]

4 levels.

First element is the average of all the elements in the original sequence.

Remaining 7 elements are fine detail in the original data.

No information has been lost.

问：怎么压缩 答：most of the value in the final level are small, fewer bits would be required to store it. this can be done by applying a threshold value the bigger the threshold the greater the loss of detail.

The coarse values by the Haar scaling function.

The fine detail by the Haar wavelet function.

此类的题往年题中有

记住每道题怎么的知识点就行.

eg2. Calculate the Haar wavelet transform for a sampled signal $s[n]=[2,1,3,-1]$ after 1 and 2 stages of the transform filterbank.

Solution:

$s[n]=[2,1,3,-1]$

First level.

$c_{1,0}=\frac{1}{\sqrt{2}}(2+1)=\frac{3}{\sqrt{2}}$

$c_{1,1}=\frac{1}{\sqrt{2}}[3+(-1)]=\frac{2}{\sqrt{2}}$

$d_{1,0}=\frac{1}{\sqrt{2}}(2-1)=\frac{1}{\sqrt{2}}$

$d_{1,1}=\frac{1}{\sqrt{2}}[3-(-1)]=\frac{4}{\sqrt{2}}$ Hence first level of the wavelet transform is $\frac{1}{\sqrt{2}}[3,2,1,4]$

second level.

$c_{0,0}=\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}(3+2)=\frac{5}{2}$

$d_{0,0}=\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}(3-2)=\frac{1}{2}$ Hence second level of the wavelet transform $\frac{1}{2}[5,1,\sqrt{2},4\sqrt{2}]$

这种题型会让你失去看书一遍. 往年题中有类似的.